

B1GMB42 Software Manual

IndianaDell workspace

2026-09-04

B1GMB42 Software Manual

Machine: Dell Precision Tower 5810 (B1GMB42)

Hostname: Tower5810

OS: Ubuntu 26.04 LTS (resolute)

Workspace: ~/Documents/IndianaDell

Companion hardware manual: B1GMB42-slot-port-inventory.md (slots, GPUs, storage, PERC, ports, lab USB dock)

Lab USB dock (WCH + CS202 audio + Genesys microSD): hardware manual USB section + docs/usb-wch-cs202-dock.md

Lab host Thumper (NVIDIA GPUs, power/clock locks): docs/thumper-gpu.md

Lab host Thumper (ZFS / disks / boot reconstruction): docs/thumper-storage.md

This manual documents every **host-facing install** the IndianaDell workspace provides: apt packages, rustup, Python venvs, built tools, Flatpak apps, GNOME preferences, Plymouth themes, optional GPU/ROCm tooling, ZFS recovery, Ventoy live persistence, and GitHub sync. Each chapter covers one topic using the same structure:

1. What gets installed
2. How it is installed
3. How to verify
4. How to customize
5. What bin/rebuild-machine does and does not do

Build PDFs: bin/build-all-docs (all manuals) or bin/build-software-manual (this book only).

Quick reference: docs/features-available.md (cheat sheet, not a replacement for this manual).

Chapter index (this manual): - 00 Front matter - 01 Introduction - 02 Rebuild and recovery - 03 Post-rebuild checklist - 04 Development - 05 Themes - 06 GPU and display - 07 GNOME session - 08 GNU Radio / SDR - 09 Ham radio - 10 HackRF Mayhem - 11 Flatpak apps - 12 Machine utilities - 13 Fac-

tory docs - 14 Gaps and limits - 15 Ventoy live session - 16 QEMU - Appendix A — Bin launchers - Appendix B — Apt packages

GitHub: <https://github.com/webaugur/IndianaDell> (private)

Supersedes: flat B1GMB42-software-inventory.md (now a stub with links here).

Chapter 1 — Introduction

What this workspace installs

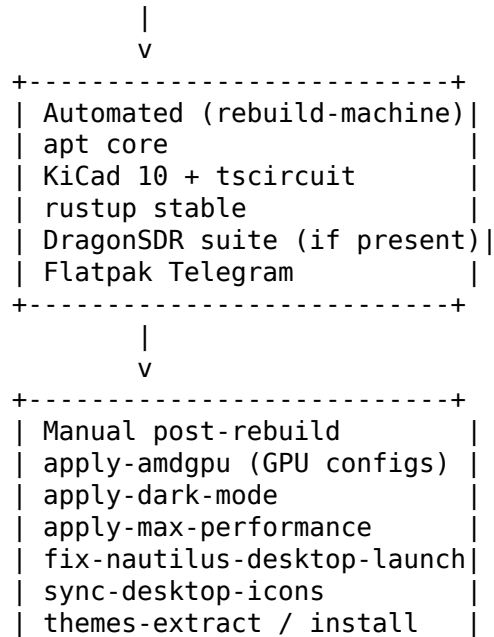
IndianaDell is a **software restoration toolkit** for Tower5810 after a fresh Ubuntu 26.04 install. It does not partition disks, configure ZFS, flash BIOS, or install Windows. It restores the development, SDR, ham radio, HackRF/Mayhem, documentation, and workstation utility stack documented in the chapters that follow.

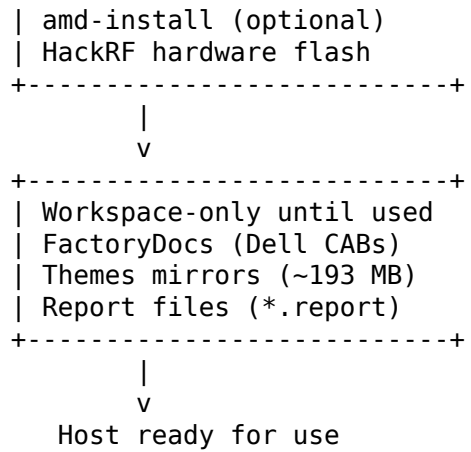
The workspace lives at ~/Documents/IndianaDell. Copy or clone it before running bin/rebuild-machine.

Install layering

Software arrives in three layers. Understanding the order prevents skipped steps after a reinstall.

Fresh Ubuntu 26.04





Automated steps run via `bin/rebuild-machine` (see Chapter 2). **Manual** steps are intentional: GPU session files, GNOME gsettings, Plymouth overlay, and hardware flashing need user context or `sudo` at the right time. **Workspace-only** content stays in the repo until you invoke the matching `bin/launcher`.

Workspace vs host paths

Location	Role
<code>~/Documents/IndianaDell/</code>	Source of truth for scripts, themes, HackRF assets
<code>/usr/</code>	Apt-installed binaries, Plymouth themes, udev rules (after apply)
<code>~/ .cargo/</code>	Rust toolchain (rustup)
<code>~/ .local/</code>	tscircuit / tsci (npm --prefix, KiCad TypeScript companion)
<code>hackrf/venv-urh/</code>	Universal Radio Hacker Python venv
<code>hackrf/build/</code>	HackRF host tools built from source
<code>hackrf/local/</code>	Optional CMAKE_INSTALL_PREFIX for built libhackrf
<code>Themes/*/mirror/</code>	Frozen copies of apt-owned theme files
<code>FactoryDocs/</code>	Dell vendor packages (not auto-installed to host)

Reading guide

If you need...	Read
Full restore after reinstall	Ch. 2 + Ch. 3
KiCad 10, tscircuit (tsc: TypeScript <code>.kicad_sch</code>)	Ch. 2 + Ch. 3 + Appendix B
Python, Rust, pandoc, Node	Ch. 4
Boot/login/desktop look	Ch. 5 + Ch. 7
FirePro GPUs, ROCm	Ch. 6
GNU Radio, gqrx, SoapySDR	Ch. 8
fldigi, WSJT-X, CHIRP	Ch. 9
HackRF, Mayhem, URH	Ch. 10
Lab USB dock (audio + microSD)	Hardware manual USB section; <code>docs/usb-wch-cs202-dock.md</code>
Telegram	Ch. 11
iotest, dellmerge	Ch. 12
Dell driver CABs	Ch. 13
Known gaps	Ch. 14
Ventoy live persistence (Uncle Wiggly + PNY rebuild stick), Grok, secrets	Ch. 15
ZFS rpool / bpool recovery	Ch. 2 + <code>docs/B1GMB42-zfs-recovery.md</code> (+ Ch. 15 live boot)
<code>/etc/default/zfs</code> force import	Ch. 2 / ZFS recovery manual — <code>ZPOOL_IMPORT_OPTS="" -f</code>
All bin/ commands	Appendix A
All apt package names	Appendix B

PATH and launchers

IndianaDell bin/ and scripts/ directories are prepended to PATH via `~/.config/indianadell/path.sh` (sourced from `~/.bashrc`). Project tools override same-named system binaries.

Related documents

- **Hardware:** `B1GMB42-slot-port-inventory.md` + PDF — GPUs, PERC, bays, ports, lab USB dock

- **Lab USB dock:** WCH hubs 1a86:8095, CS202/AB13X audio 001f:0b21, Genesys microSD 05e3:0751 — hardware manual + docs/usb-wch-cs202-dock.md (not the JMicron SATA dock)
- **ZFS recovery:** docs/B1GMB42-zfs-recovery.md + PDF — live-media rpool/bpool chroot
- **PERC IT flash:** docs/B1GMB42-perc-it-flash.md — H710 Free-DOS/Wiggly path
- **Themes deep-dive:** Themes/README.md and per-folder READMEs
- **HackRF inventory:** hackrf/MANIFEST.txt
- **Apt snapshots:** apt-full-manifest.txt, apt-hamradio-dev-manifest.txt
- **Rebuild log:** scripts/rebuild/last-run.log

Chapter 2 — Rebuild and Recovery

What gets installed

bin/rebuild-machine restores the **workstation** software stack (core apt, **KiCad 10**, rustup, Flatpak Telegram). The **SDR / ham / HackRF** stack is installed from **DragonSDR** when ~/Documents/DragonSDR is present (bin/install-dragonsdr).

How it is installed

```
cd ~/Documents/IndianaDell
chmod +x bin/* scripts/rebuild/*.sh
bin/rebuild-machine # full restore (includes DragonSDR suite if present)
bin/rebuild-machine --verify-only # check only, no installs
bin/install-dragonsdr # SDR suite alone
```

Environment overrides:

Variable	Effect
SKIP_TELEGRAM=1	Skip Flatpak Telegram install
SKIP_DRAGONSDR=1	Skip SDR suite install/verify
SKIP_HACKRF_BUILD=1	Forwarded to DragonSDR suite (skip cmake host build)
SKIP_HAM=1	Forwarded to DragonSDR (skip desktop ham apps)
SKIP_KICAD=1	Skip KiCad 10 PPA + APT_KICAD + tscircuit npm (ngspice, gerbv, 3D libs)
DRAGONSDR_ROOT=...	Override suite path (default ~/Documents/DragonSDR)

Phases (from `scripts/rebuild/rebuild-machine.sh`):

Phase	Action
1	<code>apt-get update</code>
2	Install APT_CORE — build, Python, docs, GPU utils, flatpak, gh
2b	KiCad 10 PPA + APT_KICAD + tscircuit npm (unless SKIP_KICAD=1)
3	Flatpak remote + <code>org.telegram.desktop</code> (unless skipped)
4	<code>rustup stable</code> if <code>rustc</code> missing
5	DragonSDR <code>install-suite</code> (apt SDR/ham + HackRF/Mayhem/URH) unless skipped
6	<code>chmod bin/</code> and <code>scripts</code>
7	Regenerate apt manifests; run <code>verify_stack</code>

Log file: `scripts/rebuild/last-run.log`

How to verify

```
bin/rebuild-machine --verify-only
bin/install-dragonsdr --verify-only
```

`verify_stack` checks:

- Every package in APT_CORE via `dpkg-query`
- Every package in APT_KICAD plus `kicad` / `kicad-cli` / `ngspice` / `node/npm`, `import pcbnew10.*`, and `tscircuit+@tscircuit/capacity-autorouter` (unless SKIP_KICAD=1)
- Commands: `rustc`, `cargo`, `pandoc`, `xelatex`, `vkcube`
- Launchers: `dellmerge`, `gpu-stress`, `iotest`, `apply-amdgpu`, `rebuild-machine`
- DragonSDR suite (unless SKIP_DRAGONSDR=1 or suite missing)
- Flatpak Telegram (unless SKIP_TELEGRAM=1)

Exit code 0 means all checks passed.

How to customize

- **Add workstation apt packages:** Edit `scripts/rebuild/package-lists.sh`, update Appendix B, re-run rebuild.
- **Add SDR/ham packages:** Edit `~/Documents/DragonSDR/tools/package-lists.sh`, re-run `bin/install-dragonsdr`.

- **Pin Mayhem version:** Edit DragonSDR/hackrf/scripts/download-mayhem.sh.
- **Skip heavy steps:** Use SKIP_* env vars for CI or partial recovery.

What rebuild does / does not do

Rebuild does	Rebuild does not
apt install APT_CORE + APT_KICAD (KiCad 10 PPA)	Partition disks or ZFS
tscircuit + @tscircuit/capacity-autorouter under ~/.local	sudo bin/apply-amdgpu
rustup; DragonSDR suite when present	GNOME prefs / themes by default
Flatpak Telegram	Flash HackRF / PortaPack firmware
Regenerate apt manifests	bin/amd-install (ROCm)
chmod workspace scripts	Install FactoryDocs CABs to Windows

After a successful rebuild, continue with **Chapter 3 — Post-Rebuild Checklist**.

ZFS rpool / bpool recovery

Full guide: docs/B1GMB42-zfs-recovery.md + B1GMB42-zfs-recovery.pdf (also on **DOSBOOT** IndianaDell/recovery/).

Required on the installed system: /etc/default/zfs must set force import so boot can recover after export or unclean shutdown:

```
# /etc/default/zfs
ZPOOL_IMPORT_OPTS="-f"
```

Verify anytime: `grep ZPOOL_IMPORT_OPTS /etc/default/zfs`. One-shot boot alternative: kernel cmdline `zfsforce=1`.

Quick (with scripts, from Ventoy live):

```
sudo apt-get install -y zfsutils-linux
cd ~/Documents/IndianaDell # or DOSBOOT/IndianaDell/recovery
sudo ./mount-rpool-recovery.sh mount
sudo ./scripts/recovery/mount-bpool-recovery.sh mount
```

Chapter 3 — Post-Rebuild Checklist

bin/rebuild-machine intentionally stops before steps that need a logged-in desktop, a reboot, or hardware attached. Run this checklist

once per fresh install.

1. GPU session configuration

Tower5810 has three AMD FirePro cards (W5000/W5100). Multi-GPU Wayland/X11 configs live in etc/.

```
cd ~/Documents/IndianaDell
sudo bin/apply-amdgpu
sudo reboot
```

Verify after reboot: echo \$WAYLAND_DISPLAY, glxinfo -B, vkcube on each display if needed.

See Chapter 6 for ROCm (bin/amd-install) — optional and not supported for ML on these GPUs.

```
sudo bin/apply-sensor-watch          # thermal/fan watchdog (delayed systemd)
indiana-sensor-watch --once          # print current sensors
```

2. GNOME session preferences

Run as the **desktop user** (not root):

```
bin/apply-dark-mode                  # Yaru-dark GTK, shell, icons, GDM greeter
bin/apply-max-performance            # no suspend, dimming, or night light
bin/fix-nautilus-desktop-launch      # Nautilus 50+: double-click .desktop launches
bin/sync-desktop-icons               # Nautilus 50+: show Icon= as file icon
```

Verify:

```
gsettings get org.gnome.desktop.interface color-scheme
powerprofilesctl get
bin/fix-nautilus-desktop-launch --status # expect application/x-desktop → xdg-de
bin/sync-desktop-icons -v                # sets metadata::custom-icon* from Ico
```

See Chapter 7 for every gsettings key touched, the Nautilus 50 “Allow Launching” note, and custom-icon metadata.

3. Boot splash (optional)

Default Ubuntu **bgrt** Plymouth theme shows Dell BGRT center + Ubuntu watermark. To customize:

```
bin/themes-extract                  # refresh mirrors + extract logos
# edit Themes/boot/overlay/watermark.png or background.png
sudo bin/themes-install-boot        # or --oem / --no-watermark
sudo reboot
```

Restore factory: sudo bin/themes-restore-boot

See Chapter 5.

4. HackRF hardware (when device is available)

```
bin/install-dragonsdr --verify-only    # suite present?
source bin/hackrf-env                  # PATH → ~/Documents/DragonSDR/hackrf
hackrf_info                            # should list board
bin/hackrf-flash-mayhem                # extract USB flash bundle
# follow bundle README for DFU flash
bin/hackrf-prepare-sdcard              # ensure SD tree is extracted
# copy ~/Documents/DragonSDR/hackrf/sd-card/mayhem-v2.4.0/* to FAT32 microSD root
```

Verify: hackrf_info, on-device Mayhem version, SD apps visible on PortaPack.

See Chapter 10 and ~/Documents/DragonSDR/README.md.

5. KiCad 10 and tscircuit (workstation default)

Installed by bin/rebuild-machine Phase 2b unless SKIP_KICAD=1. Opt out is rebuild-time only; this checklist just confirms the GUI/CLI stack.

KiCad 10 (PPA kicad/kicad-10.0-releases): schematic/PCB editor, 3D models, ngspice, gerbv.

tscircuit (github.com/tscircuit/tscircuit): TypeScript/React circuits via tsci. Export to KiCad, then open the files in KiCad 10. Autorouter package is @tscircuit/capacity-autorouter (tscircuit-autorouter), installed under ~/.local with tsci.

```
kicad-cli version
python3 -c 'import pcbnew; print(pcbnew.Version())'
command -v ngspice gerbv gerbview tsci
tsci --help | head
```

TypeScript → KiCad:

```
tsci init my-board
cd my-board
tsci export index.tsx -f kicad_sch    # also kicad_pcb, kicad_zip, kicad-library
```

Expect 10.0.* from kicad-cli and pcbnew. kicad-doc-id may remain on 9.x; that is not a blocker.

6. Documentation PDFs

```
bin/build-all-docs                    # software manual + hardware + inventory PDFs
# or:
bin/build-software-manual              # this manual only
```

Outputs: B1GMB42-software-manual.pdf, B1GMB42-slot-port-inventory.pdf, B1GMB42-software-inventory.pdf.

7. Machine inventory baseline

```
bin/dellmerge > blgmb42.report
sudo bin/iotest # optional storage survey
bin/gpu-stress 60 vkcube # optional GPU smoke test
```

8. FactoryDocs recovery (optional)

Only 19 of 101 pre-crash Dell packages are on disk. Re-download per FactoryDocs/README.md and MANIFEST-pre-crash.txt. These are **workspace archives**, not installed by rebuild.

See Chapter 13.

9. ZFS force import (required on this host)

After any reinstall or recovery chroot, confirm the installed system will force-import pools at boot:

```
grep '^ZPOOL_IMPORT_OPTS' /etc/default/zfs
# must show: ZPOOL_IMPORT_OPTS="-f"
```

If missing, set it in /etc/default/zfs, then `sudo update-initramfs -c -k all`. Without this, boot can hang after a recovery export or unclean shutdown. See Chapter 2 and docs/B1GMB42-zfs-recovery.md.

Quick verification block

```
cd ~/Documents/IndianaDell
bin/rebuild-machine --verify-only
kicad-cli version # expect 10.0.*
tsci --help >/dev/null && echo OK tscircuit # TypeScript → KiCad
grep '^ZPOOL_IMPORT_OPTS' /etc/default/zfs # expect "-f"
source bin/hackrf-env
. ~/.cargo/env && rustc --version
gnuradio-config-info --version
bin/urh --version
```

Summary table

Step	Command	Reboot?
GPU configs	sudo bin/apply-amdgpu	Yes
Dark mode	bin/apply-dark-mode	No

Step	Command	Reboot?
Max performance	bin/apply-max-performance	No
Nautilus 50 .desktop launch	bin/fix-nautilus-desktop-launch	No
Nautilus 50 .desktop icons	bin/sync-desktop-icons	No
Custom boot	sudo bin/themes-install-boot	Yes
HackRF flash	bin/hackrf-flash-mayhem + DFU	Maybe
KiCad 10 + tscircuit	kicad-cli version; tsci export ... -f kicad_sch	No
ROCM (optional)	bin/amd-install	Yes
All doc PDFs	bin/build-all-docs	No
ZFS force import	check /etc/default/zfs □ ZPOOL_IMPORT_OPTS="-f"	If initramfs updated
Ventoy persistence seed	~/bin/seed-ventoy-persistence.sh	No

Chapter 4 — Development Toolchain

What gets installed

Component	Version / path	Install source
Python pip, venv	3.14 (system) apt	Ubuntu base + apt python3-pip, python3-venv
numpy, scipy, matplotlib PyQt5	apt apt	science stack
Rust C/C++ build	1.96.1 stable apt	GNU Radio Companion, some tools rustup to ~/.cargo/bin
Clang/LLVM	apt	build-essential, cmake, pkg-config
SSL/USB/FFTW/Volk apt dev libs		bindgen-style Rust, native tooling SDR and native builds

Component	Version / path	Install source
ARM cross-compile	apt	PortaPack Mayhem firmware (gcc-arm-none-eabi)
pandoc + XeLaTeX	apt	Manual PDF generation
Git, curl, wget	apt	repos and downloads
GitHub CLI (gh)	2.46 (apt)	HTTPS/token fallback; gh auth login for API access
Node.js + npm	apt (APT_KICAD)	tscircuit CLI
tscircuit (tsci)	npm --prefix ~/.local	TypeScript/React <code>□</code> .kicad_sch / .kicad_pcb (Chapter 3)
@tscircuit/capacitor	npm --prefix ~/.local	PCB autorouter used by tscircuit

Python bindings verified on this host: gnuradio, SoapySDR, Hamlib (capital H in Python).

Project venv: URH lives in ~/Documents/DragonSDR/hackrf/venv-urh/ (Chapter 10), not system-wide.

How it is installed

Apt (rebuild Phase 2): packages listed in Appendix B under “Development” and shared dev libraries.

Rust (rebuild Phase 5):

```
curl --proto '=https' --tlsv1.2 -sSf https://sh.rustup.rs | sh -s -- -y --default-source ~/.cargo/env
```

Rebuild skips rustup if rustc is already on PATH.

How to verify

```
python3 --version
python3 -c "import numpy, scipy, matplotlib; print('OK')"
. ~/.cargo/env && rustc --version && cargo --version
cmake --version | head -1
arm-none-eabi-gcc --version | head -1
pandoc --version | head -1
xelatex --version | head -1
node --version && npm --version
tsci --help >/dev/null && echo OK tscircuit
```

How to customize

- **New Python project venv:** `python3 -m venv myproject/.venv`
&& `. myproject/.venv/bin/activate`
- **Rust toolchain:** `rustup toolchain install stable`, `rustup component add clippy`
- **Per-project Rust SDR crates:** `cargo add rtlsdr` etc. — no workspace scaffold ships with IndianaDell
- **Manual PDFs:** `bin/build-all-docs` (all docs) or `bin/build-software-manual` (software manual only)

What rebuild does / does not do

Does	Does not
Install all dev apt packages	Create application-specific venvs beyond URH
Install rustup if missing	Pin a non-stable Rust toolchain
chmod workspace scripts	Install IDE or editor plugins

Chapter 5 — Themes (Boot, Login, Desktop)

What gets installed

The **Themes/** module (~194 MB with mirrors) documents and customizes three visual stages:

Stage	What you see	Apt packages	Workspace folder
Boot	Dell BGRT center + spinner + Ubuntu watermark	plymouth, plymouth-theme-spinner, ...	Themes/boot/
Login	GDM on GNOME Shell	gdm3, gnome-shell, yaru-theme-gnome-shell	Themes/login/
Desktop	Yaru GTK, icons, shell	yaru-theme-gtk, yaru-theme-icon, ...	Themes/desktop/

Active Plymouth theme: `bgrt at /usr/share/plymouth/themes/bgrt/bgrt.plymouth`
Extracted boot logos:

- Themes/boot/extracted/bgrt-firmware-oem.png — Dell from UEFI BGRT (/sys/firmware/acpi/bgrt/image)
- Themes/boot/extracted/ubuntu-watermark-dark.png — bottom Ubuntu text

Custom Plymouth install target: indianadell theme under /usr/share/plymouth/themes/indianadell

Each subfolder has its own README.md and apt-packages.txt. See Themes/MANIFEST.txt for a one-page map.

How it is installed

Themes are **not** applied by bin/rebuild-machine. Use launchers:

```
bin/themes-extract           # snapshot apt-owned files + extract logos (~193 MB)
sudo bin/themes-install-boot # install custom Plymouth from boot/overlay/
sudo bin/themes-restore-boot # revert to stock bgrt
bin/apply-dark-mode          # login + desktop dark (Chapter 7)
```

Boot overlay workflow:

```
cp my-logo.png Themes/boot/overlay/watermark.png # bottom Ubuntu text only
sudo bin/themes-install-boot
```

```
cp my-splash.png Themes/boot/overlay/background.png
sudo bin/themes-install-boot --oem # replace center (disable BGRT)
```

```
sudo bin/themes-install-boot --no-watermark # hide bottom logo
```

Every Plymouth install runs update-initramfs -u — **reboot required** to preview.

How to verify

```
plymouth-set-default-theme -l
plymouth-set-default-theme # shows active theme
ls Themes/boot/extracted/
gsettings get org.gnome.desktop.interface gtk-theme # after apply-dark-mode
```

How to customize

Deep documentation lives in module READMEs — this chapter summarizes:

- **Boot:** Themes/boot/README.md — overlay/, stock/, indianadell/ plymouth definition
- **Login:** Themes/login/README.md — GDM greeter assets mirrored from apt
- **Desktop:** Themes/desktop/README.md — GTK/icon/shell Yaru variants

Center Dell logo comes from BIOS BGRT firmware, not an Ubuntu file. Change it in BIOS setup or use `--oem` with your PNG.

What rebuild does / does not do

Does	Does not
Install plymouth, gdm3, Yaru via apt (Phase 2-3)	Run themes-extract or themes-install-boot Set dark mode or GDM greeter prefs Copy overlay PNGs into initramfs

Run Chapter 3 checklist items 2 and 3 after rebuild.

Chapter 6 — GPU and Display

What gets installed

Component	Source	Purpose
vulkan-tools, mesa-utils, clinfo etc/ multi-GPU configs	apt (APT_CORE) workspace	Vulkan/OpenGL/OpenCL diagnostics Wayland, X11, udev, GDM tweaks
amd-radeon/ scripts	workspace	Optional ROCm driver install
bin/gpu-stress	workspace	3-GPU Vulkan smoke test

Hardware (this machine): 2x AMD FirePro W5000 + 1x FirePro W5100. Vulkan and OpenCL work for graphics/compute smoke tests. **ROCm ML/HIP is not supported** on these cards (see Chapter 14).

How it is installed

Apt (automated): GPU utility packages install during rebuild Phase 2.

Session configs (manual):

```
sudo bin/apply-amdgpu    # runs etc/apply.sh
sudo reboot
```

etc/apply.sh installs:

- etc/environment.d/99-amdgpu-wayland.conf

- etc/X11/xorg.conf.d/20-amdgpu-multi-gpu.conf
- etc/modprobe.d/amdgpu-multigpu.conf (runpm=0)
- etc/udev/rules.d/99-amdgpu-multigpu.rules (tags + DPM auto hook)
- etc/amdgpu-set-dpm-auto.sh □ /usr/local/sbin/indiana-amdgpu-dpm-auto
- etc/profile.d/amdgpu-multigpu.sh
- etc/gdm3/custom.conf (if present)

DPM auto (all cards): every amdgpu uses load-based clocks (auto / balanced). These FirePro W5000/W5100 overheat when pinned to high. apply-amdgpu runs the helper immediately; udev re-applies when cards appear at boot.

```
# verify
for c in /sys/class/drm/card[0-9]/device; do
    [[ -f $c/power_dpm_force_performance_level ]] || continue
    echo "$(basename $(dirname $c)): level=$(cat $c/power_dpm_force_performance_level)
done
# expect: level=auto state=balanced on card1..card3
```

Do not set high on this machine. To force clocks only for a short smoke test:
echo high | sudo tee .../power_dpm_force_performance_level,
then echo auto when done.

Optional ROCm:

```
bin/amd-preflight      # check prerequisites
bin/amd-install        # full driver stack from amd-radeon/
bin/amd-verify
bin/amd-uninstall     # remove if needed
```

How to verify

```
vkcube                # Vulkan cube (per display)
clinfo | head -30      # OpenCL platforms/devices
glxinfo -B             # OpenGL renderer
bin/gpu-stress 60 vkcube # stress all GPUs ~60s
lspci -nn | grep -i vga
```

How to customize

- Edit files under etc/ before re-running sudo bin/apply-amdgpu
- ROCm install scripts and README in amd-radeon/ — machine-specific; read preflight output
- Hardware details: B1GMB42-slot-port-inventory.md Video section

What rebuild does / does not do

Does	Does not
Install mesa-utils, vulkan-tools, clinfo Ensure bin/gpu-stress is executable	Run apply-amdgpu Install ROCm Configure monitor layout (use GNOME Settings)

Required post-rebuild: `sudo bin/apply-amdgpu` + reboot (Chapter 3).

Sensor watchdog

thermald does not run on this Haswell Xeon. `bin/apply-sensor-watch` installs `indianadell-sensor-watch.service` (starts after `graphical.target`, sleeps 120 s on a cold boot, then polls). It logs out-of-range CPU/GPU/DIMM/fan/SMART temps to syslog, pops a GNOME notification on the logged-in session bus, and steps GPU clocks / CPU `scaling_max_freq` down until under 80 °C (back up after a 72 °C dead-band). Disk SMART uses `smartctl -n standby,0` so sleeping HDDs are not woken.

A desktop notice is sent on **systemctl reload** and **systemctl restart**, not on first start or stop. Test the path without cycling the unit:

```
sudo indiana-sensor-watch --notify test
sudo systemctl reload indianadell-sensor-watch
sudo bin/apply-sensor-watch
indiana-sensor-watch --once
journalctl -t indiana-sensor-watch -f
```

Config: `/etc/indiana-sensor-watch.conf` (defaults in `etc/indiana-sensor-watch.conf`). `THROTTLE=0` / `THROTTLE_CPU=0` disable clock writes. `dell_smm fan2` is ignored (unused `HDD_FAN` header).

Monitor input select

These FirePros are mini-DP only. There is no `/dev/cec`, so HDMI-CEC cannot switch the set.

`indiana-monitor-input` uses **DDC/CI** (`ddcutil`, VCP 60) on the current cable:

```
indiana-monitor-input detect
indiana-monitor-input grab    # request DisplayPort-1 (this PC)
indiana-monitor-input set hdmi1
```

Verified on this Samsung (2018) via `card2-DP-4` / `i2c-7`: EDID works, **I2C 0x37 does not answer**. Until the OSD “DDC/CI” option is on (or a different

panel is used), grab will fail with that fact. User must be in group i2c (or use sudo).

IR HDMI select (Arduino Uno)

When DDC/CEC cannot switch the TV, a 940 nm LED on an Uno can send one Samsung discrete code. Wiring and flash: `tools/ir-blaster/README.md`.

```
indiana-ir-flash                # arduino-cli; no IDE
indiana-ir-send hdmi2
```

Host waits for READY/OK on the serial fd (Uno resets on port open). Group dialout required.

Lab host note — Thumper (NVIDIA)

This chapter is **Tower5810 / AMD only**. The lab host **thumper.local** (Dell Precision T5610) runs **NVIDIA TITAN Xp** and is **not** configured with `apply-amdgpu`.

Full Thumper GPU doc (inventory, dual-card plan, **power/clock locking**, LingBot-Map pointer):

□ `docs/thumper-gpu.md`

Summary: a second TITAN Xp with a weak cooler can be limited with `nvidia-smi -pl / -lgc` (and a oneshot systemd unit). Prefer `CUDA_VISIBLE_DEVICES` so heavy jobs stay on the well-cooled card. DragonSDR LingBot-Map lives under `~/Data/lingbot-map` on Thumper — see also `~/Documents/DragonSDR/tools/lingbot-map/README.md`.

Chapter 7 — GNOME Session

What gets installed

GNOME desktop preferences applied via `gsettings` (user session) and optionally GDM (login greeter). No apt packages beyond what Ubuntu desktop already provides (`gdm3`, `gnome-shell`, Yaru themes).

Scripts:

- `scripts/gnome/apply-dark-mode.sh` via `bin/apply-dark-mode`
- `scripts/gnome/apply-max-performance.sh` via `bin/apply-max-performance`
- `scripts/gnome/fix-nautilus-desktop-launch.sh` via `bin/fix-nautilus-desktop-launch`

- `scripts/gnome/sync-desktop-icons.sh` via `bin/sync-desktop-icons`

How it is installed

Run as the **logged-in desktop user**, not root:

```
bin/apply-dark-mode
bin/apply-max-performance
bin/fix-nautilus-desktop-launch # Nautilus 50+ .desktop double-click launch
bin/sync-desktop-icons         # Nautilus 50+ show Icon= as file icon
```

apply-dark-mode

Sets:

Schema	Key	Value
org.gnome.desktop.interface	color-scheme	prefer-dark
org.gnome.desktop.interface	gtk-theme	Yaru-dark (override: GTK_THEME=)
org.gnome.desktop.interface	icon-theme	Yaru-dark
org.gnome.shell.ubuntu	color-scheme	prefer-dark
org.gnome.desktop.wm.preferences	theme	Yaru-dark
org.gnome.settings-daemon.plugins.color	night-light	disabled

With `APPLY_GDM=1` (default), also sets GDM greeter dark via `sudo dbus-run-session gsettings ....`

apply-max-performance

Sets:

- Power plugin: no suspend on AC/battery, no idle dim, lid close does nothing
- Session idle delay: 0
- Screensaver: no blanking or lock on idle
- Night light: off
- `powerprofilesctl set performance` when available

Nautilus 50 — “Allow Launching” removed

Change (GNOME Files / Nautilus 50, Ubuntu 26.04): Nautilus no longer runs `FreeDesktop .desktop` files itself. The old “Allow Launching” / `trusted-launcher` path is gone (security). Double-click falls through to

the default handler for MIME type application/x-desktop, which is often a text editor (gedit / gnome-text-editor). So double-clicking SDRPlusPlus.desktop (or any app launcher on disk) **edits** the file instead of **starting** the app.

Fix: register a small MIME handler that launches the entry via gio launch:

Piece	Path
Handler app	~/.local/share/applications/xdg-desktop-launch.desktop
Wrapper script	~/.local/bin/xdg-desktop-launch
MIME default	application/x-desktop ☐ xdg-desktop-launch.desktop

Double-click path after install: **Files** ☐ **xdg-open** ☐ **wrapper** ☐ **gio launch** ☐ **your app**.

```
bin/fix-nautilus-desktop-launch # install / reinstall
bin/fix-nautilus-desktop-launch --status # check MIME + files
bin/fix-nautilus-desktop-launch --uninstall # remove handler
```

Portable: the script is self-contained. Copy scripts/gnome/fix-nautilus-desktop-launch.sh to any Ubuntu/GNOME box and run as the desktop user (no root, no IndianaDell tree required).

CLI verify:

```
xdg-mime query default application/x-desktop # expect xdg-desktop-launch.desktop
xdg-open /path/to/App.desktop # should start the app
~/.local/bin/xdg-desktop-launch /path/to/App.desktop
```

Nautilus 50 — generic .desktop icons

Change: even after launch works, Nautilus 50+ still shows the generic application-x-desktop MIME icon in the file view. It **ignores** the FreeDesktop Icon= field on the .desktop file for the list/icon view. Icons in the GNOME Shell app grid (from XDG application menus) are a different path and usually look correct; this problem is about **Files** showing launcher files as plain documents.

Fix: set GIO metadata that Nautilus still honors:

Attribute	When used	Example
metadata::custom-icon	Icon= is an absolute or relative file path	file:///home/user/Applications/sdrpp.pr

Attribute	When used	Example
metadata::custom-icon-name	Icon= is a theme icon name	utilities-terminal

The script reads the first Icon= under [Desktop Entry] only (not Icon[lang]=, not other groups), then:

1. Absolute path (/...) → metadata::custom-icon with a file:// URI; clear custom-icon-name
2. Relative path (foo/bar.png) → resolve relative to the .desktop file's directory; same as absolute
3. Theme name (no /) → metadata::custom-icon-name; clear custom-icon
4. Missing icon file → warning and skip (does not fail the whole run)

Usage

```
bin/sync-desktop-icons           # scan default directories
bin/sync-desktop-icons -v        # log each set / skip / rename
bin/sync-desktop-icons --dry-run # print actions, no gio set / rename
bin/sync-desktop-icons --file PATH # one .desktop (inotify-friendly)
bin/sync-desktop-icons --dir DIR  # add/replace scan dir (repeatable)
bin/sync-desktop-icons --watch    # inotify loop (needs inotify-tools)
bin/sync-desktop-icons --clear-missing # unset custom-icon* if Icon= absent
bin/sync-desktop-icons --no-rename  # keep chrome-*-Default.desktop names
```

Default scan directories (when --dir is not used and SYNC_DESKTOP_ICON_DIRS is unset):

- \$HOME/.local/share/applications
- \$HOME/Applications
- \$HOME/Desktop

Only **top-level** *.desktop files in each directory are processed (flat XDG apps layout and a personal Applications / Desktop folder). Nested trees are not walked.

Rename Chrome gibberish basenames

Chrome/Chromium PWAs create launchers like chrome-lodlkdfmihgonocnmdehnfgiljnadcf-Default.desktop while Name= is a short human label (X, YouTube). **By default** (--rename), if the file **and** its directory are **writable**, matching basenames are renamed to \${Name}.desktop before icon metadata is applied.

Rule	Behavior
Pattern	chrome-<id>-Default.desktop, chrome-<id>.desktop (id ∈ 16 alnum); same for chromium-
Source of new name	First Name= under [Desktop Entry] only (not action groups, not Name[lang]=)
Not writable	Skip rename, warn, still try icon metadata
Target already exists	Skip rename, warn (no overwrite)
Disable	--no-rename

Example: ~/Desktop/chrome-lodlk...-Default.desktop (Name=X) →
~/Desktop/X.desktop.

Environment:

Variable	Effect
SYNC_DESKTOP_ICON_DIRS	Colon-separated directory list (overrides defaults when set)

Dependencies: gio (libglib2.0-bin, already on Ubuntu desktop). --
watch also needs inotifywait (inotify-tools).

Portable / PATH install: bin/sync-desktop-icons resolves its own path
with readlink -f, so a symlink from ~/.local/bin/sync-desktop-
icons into the repo still finds scripts/gnome/. The script itself needs only
gio; copy scripts/gnome/sync-desktop-icons.sh alone if you want it
off-tree.

Companion: run **after** bin/fix-nautilus-desktop-launch so double-
click starts the app **and** the file view shows the right icon. Neither script
replaces the other.

CLI verify:

```
bin/sync-desktop-icons -v
# pick a launcher you care about:
gio info -a metadata::custom-icon -a metadata::custom-icon-name \
"$HOME/Applications/SomeApp.desktop"
# open Files on that folder – icon should match Icon=
```

Watch mode example (re-apply when a launcher is saved or dropped into
a scan dir):

```
bin/sync-desktop-icons --watch -v
# or a single-file handler:
inotifywait -m -e close_write,moved_to,create --include '\.desktop$' \
"$HOME/Applications" | while read -r dir _ file; do
```

```
bin/sync-desktop-icons --file "$dir$file"
done
```

How to verify

```
gsettings get org.gnome.desktop.interface color-scheme
gsettings get org.gnome.desktop.interface gtk-theme
gsettings get org.gnome.settings-daemon.plugins.power sleep-inactive-ac-type
powerprofilesctl get
bin/fix-nautilus-desktop-launch --status
bin/sync-desktop-icons -v
```

Log out and back in to confirm GDM greeter if dark mode was applied.
Refresh or reopen the Files window after sync-desktop-icons if icons do not update immediately.

How to customize

- Override themes: `GTK_THEME=Yaru-viridian bin/apply-dark-mode`
- Skip GDM: `APPLY_GDM=0 bin/apply-dark-mode`
- Revert individual keys with `gsettings reset ...` or GNOME Settings app
- Login/desktop asset mirrors: Themes/login/, Themes/desktop/
- Remove .desktop launch fix: `bin/fix-nautilus-desktop-launch --uninstall`
- Extra icon scan dirs: `SYNC_DESKTOP_ICON_DIRS="$HOME/Apps:$HOME/Desktop"`
`bin/sync-desktop-icons`
- Clear stale custom icons when Icon= is gone: `bin/sync-desktop-icons --clear-missing`
- Unset metadata on one file:
`gio set -t unset PATH metadata::custom-icon`
`gio set -t unset PATH metadata::custom-icon-name`

What rebuild does / does not do

Does	Does not
Install gdm3, gnome-shell, Yaru via apt	Change any gsettings Set performance power profile Install the Nautilus 50 .desktop MIME handler Run sync-desktop-icons or set custom-icon metadata

Run dark mode, max performance, fix-nautilus-desktop-launch, and sync-desktop-icons after every fresh install (Chapter 3).

Chapter 8 — GNU Radio and Desktop SDR

Ownership

GNU Radio, SoapySDR, and desktop SDR apps are installed by the **DragonSDR** suite, not by IndianaDell apt lists.

Item	Location
Suite install	~/Documents/DragonSDR/bin/install-suite
Package list	~/Documents/DragonSDR/tools/package-lists.sh (APT_SDR)
IndianaDell wrapper	bin/install-dragonsdr

What gets installed

Component	Packages / path
GNU Radio Companion blocks	gnuradio, gnuradio-dev, gnuradio-doc gr-osmosdr, gr-limesdr, gr-fosphor, gr-air-modes, gr-hpsdr, gr-dab, gr-satellites
SoapySDR	libsoapysdr-dev, python3-soapysdr, modules
Hardware libs	RTL-SDR, HackRF, Airspy, bladeRF, Lime, UHD
Desktop apps	gqrx-sdr, quisk, inspectrum, hacktv

SoapySDR modules (typical host): HackRF, RTL-SDR (osmosdr), Airspy, bladeRF, Lime, MiriSDR, HydraSDR, PlutoSDR, Red Pitaya, remote, audio, UHD.

How it is installed

```
bin/install-dragonsdr                # full suite (apt + HackRF workspace)
# or only packages:
bin/install-dragonsdr --apt-only
```

Called automatically during bin/rebuild-machine when ~/Documents/DragonSDR is present (SKIP_DRAGONSDR=1 to skip).

Typical workflow:

```
g++ myflowgraph.grc                 # compile Companion graph
gqrx                                 # general receiver GUI
```

```
quisk                                # transceiver GUI
inspectrum capture.cf32             # visualize IQ files
```

For HackRF-specific host tools and URH, see Chapter 10.

How to verify

```
bin/install-dragonsdr --verify-only
gnuradio-config-info --version
python3 -c "import SoapySDR; print('SoapySDR OK')"
```

SoapySDRUtil --info

```
command -v gqrx
```

Chapter 9 — Ham Radio (Desktop)

Ownership

Desktop ham applications are part of the **DragonSDR** suite (APT_HAM in tools/package-lists.sh).

What gets installed

Application	Command	Apt package	Role
fldigi	fldigi	fldigi	Digital modes (PSK, RTTY, ...)
WSJT-X	wsjtx	wsjtx, wsjtx-data	FT8, JT65, weak-signal
CHIRP	chirpw, chirpc	chirp	Radio programming
direwolf	direwolf	direwolf	Sound-card TNC / APRS
gpredict	gpredict	gpredict	Satellite pass prediction
grig	grig	grig	Hamlib rig control GUI

Application	Command	Apt package	Role
xastir	xastir	xastir, xastir-data	APRS map client
Hamlib	API	libhamlib-dev, libhamlib-utils, python3-hamlib	Rig control library

How it is installed

```
bin/install-dragonsdr
# omit ham apps:
SKIP_HAM=1 bin/install-dragonsdr
```

xastir debconf: suite install preseeds xastir/install-setuid boolean false to avoid interactive hangs.

```
fldigi &
wsjtx &
chirpw &
direwolf -p
gpredict &
grig &
xastir &
```

How to verify

```
command -v fldigi wsjtx chirpw direwolf gpredict grig xastir
python3 -c "import Hamlib; print('Hamlib OK')"
```

```
bin/install-dragonsdr --verify-only
```

Configure rig control in each app via Hamlib model selection.

Chapter 10 — HackRF and PortaPack Mayhem

Ownership

The HackRF / PortaPack Mayhem workspace moved out of IndianaDell into **DragonSDR**:

Item	Path
Workspace	~/Documents/DragonSDR/hackrf/
Manifest	~/Documents/DragonSDR/hackrf/MANIFEST.txt

Item	Path
Suite install	~/Documents/DragonSDR/bin/install-suite
IndianaDell wrappers	bin/hackrf-*, bin/urh, bin/install-dragonsdr

What gets installed

Recommended firmware: PortaPack Mayhem v2.4.0

Apt packages

hackrf, hackrf-firmware, libhackrf-dev, hackrf-doc, inspectrum, hacktv, dfu-util, openocd, ARM GCC toolchain, plus GNU Radio/SoapySDR deps (Chapter 8).

Built from source (DragonSDR/hackrf/build/)

Tool	Notes
hackrf_sweep	Spectrum sweep
hackrf_info, hackrf_transfer, ...	Host utilities
libhackrf.so	Under hackrf/build/libhackrf/src/

Install prefix: hackrf/local/ (CMAKE_INSTALL_PREFIX).

Release assets (hackrf/releases/)

File	Purpose
FIRMWARE_mayhem_v2.4.0.zip	USB flash bundle
COPY_TO_SDCARD_hackrf_mayhem_v2.4.0	PortaPack microSD
no-world-map.zip	no-world-map
OCI_hackrf_mayhem_v2.4.0.webflash	Web flasher image

Extracted SD tree: hackrf/sd-card/mayhem-v2.4.0/

Source repos (hackrf/repos/)

hackrf, mayhem-firmware (+ submodules), portapack-hackrf, urh, hacktv

Python venv

hackrf/venv-urh/ — Universal Radio Hacker. Launch: bin/urh (wrapper
□ DragonSDR).

udev

hackrf/scripts/99-hackrf.rules □ /etc/udev/rules.d/

How it is installed

```
bin/install-dragonsdr                # full suite
SKIP_HACKRF_BUILD=1 bin/install-dragonsdr
bin/install-dragonsdr --hackrf-only  # workspace only (apt already done)
```

Manual (hardware present):

```
source bin/hackrf-env                # PATH → DragonSDR/hackrf/build
bin/hackrf-flash-mayhem
bin/hackrf-prepare-sdcard
bin/hackrf-build-mayhem              # compile Mayhem from source
bin/hackrf-download-mayhem
bin/urh
```

How to verify

```
bin/install-dragonsdr --verify-only
source bin/hackrf-env
hackrf_info
ls ~/Documents/DragonSDR/hackrf/sd-card/mayhem-v2.4.0/APPS | wc -l
bin/urh --version
```

Host card reader (lab USB dock)

IndianaDell machines share a USB 2 dock with a **Genesys Logic 05e3:0751** microSD slot (enumerates only with a card in). On 2026-09-04 a PortaPack Mayhem card (FAT32 label **HackRF**, UUID 300C-16B0) auto-mounted at /run/media/user/HackRF.

Use /dev/disk/by-id/usb-Generic_STORAGE_DEVICE-0:0, not sdd. The reader sat on a USB 2 daisy-chain; the kernel logged a dirty FAT volume and a brief disconnect on write. Run fsck.vfat before putting that card back in a PortaPack. Dock IDs, audio jack, and topology are in the **hardware manual** USB section (B1GMB42-slot-port-inventory.md) and docs/usb-wch-cs202-dock.md.

bin/hackrf-prepare-sdcard still writes a prepared tree; this dock is just the host slot to mount or image that card.

Chapter 11 — Flatpak Applications

What gets installed

App	Flatpak ID	Version (this host)
Telegram	org.telegram.desktop	6.9.3

Runtime dependency: flatpak package from APT_CORE.

How it is installed

Rebuild Phase 4:

```
flatpak remote-add --if-not-exists flathub https://dl.flathub.org/repo/flathub.flatpakrepo
flatpak install -y flathub org.telegram.desktop
Skip with SKIP_TELEGRAM=1 bin/rebuild-machine.
```

How to verify

```
flatpak list --app | grep telegram
flatpak run org.telegram.desktop --version 2>/dev/null || true
bin/rebuild-machine --verify-only
```

How to customize

```
flatpak update org.telegram.desktop
flatpak override --user org.telegram.desktop ... # permissions, env
bin/apply-dark-mode sets prefer-dark color scheme; Flatpak GTK4
apps pick this up via portal when supported.
```

What rebuild does / does not do

Does	Does not
Install flatpak via apt	Install SDRangel, SigDigger (not on Flathub)
Add flathub remote + Telegram	Pin Telegram to a specific commit
Treat Telegram miss as non-fatal on install (warns in log)	Install other Flatpak apps by default

Chapter 12 — Machine Utilities

What gets installed

Workspace scripts for inventory, storage benchmark, and GPU stress — no dedicated apt packages beyond shared GPU utils (mesa-utils, vulkan-tools).

Utility	Launcher	Script	Output
Dell inventory	bin/dellmerge	scripts/dell/dellmerge.sh	dellmerge.sh *.report files
Storage survey	bin/iotest	scripts/storage/iotest.sh	iotest.sh (sudo)
GPU stress	bin/gpu-stress	scripts/gpu/gpu-stress.sh	Vulkan/EGL per GPU
EFI / BIOS timing	bin/efi-timing-suite	scripts/efi/efi-timing-suite.sh	B1GMB42.timing (sudo)

Example reports in workspace: blgmb42.report, B1GMB42.ioperf, B1GMB42.timing (from prior runs).

How it is installed

Scripts ship with the workspace. Rebuild Phase 10 runs `chmod +x` on `bin/*` and `scripts/*/*.sh`.

```
bin/dellmerge > blgmb42.report
```

```
sudo bin/iotest
```

```
bin/gpu-stress 60 vkcube
```

```
sudo bin/efi-timing-suite # before/after BIOS A/B changes
```

How to verify

```
[[ -x bin/dellmerge && -x bin/iotest && -x bin/gpu-stress && -x bin/efi-timing-suite ] && \
bin/rebuild-machine --verify-only # checks dellmerge, gpu-stress, iotest, apply
head -20 blgmb42.report 2>/dev/null || bin/dellmerge | head -20
```

How to customize

- Edit `scripts/dell/dellmerge.sh` to add inventory fields
- `gpu-stress` accepts duration and backend (vkcube default)
- `iotest` targets block devices — read script header before running on production pools

- `efi-timing-suite` writes a machine-local timing baseline; re-run after BIOS changes for A/B compare

What rebuild does / does not do

Does	Does not
chmod utility launchers	Run <code>dellmerge</code> , <code>iotest</code> , or <code>efi-timing</code> automatically
Verify launcher executables exist	Archive reports to a fixed path

Chapter 13 — FactoryDocs (Workspace Archive)

What gets installed

Nothing on the Linux host automatically. FactoryDocs is a sorted archive of Dell T5810 vendor support packages for Windows recovery, firmware, and reference — stored only in the workspace.

Metric	Pre-crash	Current
Sorted packages	101	19
GPU drivers (FirePro/Quadro)	Yes	Missing
PERC H710 driver/firmware	Yes	Missing
Audio / input drivers	Yes	Missing
Win7/10/WinPE CAB packs	Yes	Missing

Full pre-crash file list: `FactoryDocs/MANIFEST-pre-crash.txt` (91 items flagged MISS).

Layout

Folder	Contents
System-T5810/	BIOS, chipset, ME, TPM, manuals
GPU/	AMD FirePro, NVIDIA (empty — re-download)
Storage/	PERC H710, Intel RST, SSD/HDD firmware
Network/	Intel Ethernet
Dell-Management/	Command Update, Configure
Expansion-Cards/	Serial, Thunderbolt docs
_Misc/	Non-Dell packages

Folder	Contents
_incoming/	Drop new Dell downloads here

How it is installed

Ingest new downloads:

```
cp ~/Downloads/* ~/Documents/IndianaDell/FactoryDocs/_incoming/
python3 ~/Documents/IndianaDell/FactoryDocs/_sort_factory_docs.py
```

Priority re-downloads from Dell T5810 drivers:

1. GPU/AMD-FirePro/Windows/ — Video_Driver_C5FPW (W5000/W5100)
2. Storage/RAID-Controller-PERC/Windows/ — PERC H710
3. System-T5810/Windows-10/ — T5810-win10 CAB
4. Audio/Windows/ — Audio_Driver_5P33P
5. Dell-Management/Windows/ — Command Configure, Monitor

Windows installs use Dell CAB/EXE packages from these folders — not apt.

How to verify

```
find FactoryDocs -type f ! -path '*/_incoming/*' | wc -l
cat FactoryDocs/README.md
grep MISS FactoryDocs/MANIFEST-pre-crash.txt | wc -l
```

How to customize

- _sort_factory_docs.py dedupes and sorts by hardware category
- Cross-reference hardware manual: BIGMB42-slot-port-inventory.md for what hardware needs drivers

What rebuild does / does not do

Does	Does not
Nothing	Copy FactoryDocs to system paths Install Windows drivers Download missing CABs

FactoryDocs recovery is a **manual** ongoing task (Chapter 3 item 7).

Chapter 14 — Gaps and Limits

Documented boundaries of what IndianaDell does **not** install or support on this host.

Not installed

Item	Notes
SDRangel, SigDigger	Not on Flathub; use gqrx + URH + inspectrum
Rust SDR crate workspace	Toolchain only — add crates per project with <code>cargo add</code>
ZFS / disk layout tools	Out of scope — handled at OS install time
Windows / dual-boot	FactoryDocs holds CABs; no auto-install
HackRF hardware test	No device attached at last verify (2026-07-05)
Ventoy full seed on every boot	Manual — run <code>~/bin/seed-ventoy-persistence.sh</code> after changes
PNY rebuild-stick seed automation	Manual loop-mount of <code>ubuntu-26.04.dat</code> (see Ch. 15); stick units only fix groups/SSH/hostname
Freerouting / pcb2gcode	Not installed — tscircuit ships <code>@tscircuit/capacity-autorouter</code>
kicad-doc-id 10.x	Universe still 9.0.8; omitted from <code>APT_KICAD</code>

Lost in TPM/ZFS crash

Item	Recovery
Pre-crash full apt list	Not found — use <code>apt-full-manifest.txt</code> from current rebuild
Pre-crash FactoryDocs	82/101 packages missing — see <code>MANIFEST-pre-crash.txt</code>
Pre-crash apt-hamradio list	Regenerated by <code>rebuild save_manifests()</code>

GPU / ROCm matrix

Capability	W5000 / W5100 on Ubuntu 26.04
Desktop amdgpu	Yes — with etc/ configs
Vulkan (vkcube)	Yes
OpenCL (clinfo)	Yes (limited)
ROCm HIP / ML training	No — not in AMD ROCm support matrix for FirePro W5000/W5100

Use `bin/amd-preflight` before `bin/amd-install`; expect warnings for these GPUs.

Rebuild intentional omissions

These remain **manual** by design (see Chapter 3):

- `sudo bin/apply-amdgpu`
- `bin/apply-dark-mode`, `bin/apply-max-performance`, `bin/fix-nautilus-desktop-launch`, `bin/sync-desktop-icons`
- `bin/themes-extract`, `sudo bin/themes-install-boot`
- `bin/hackrf-flash-mayhem` (DFU with hardware)
- `bin/amd-install` (optional ROCm)

Future enhancement could fold GNOME/theme steps into rebuild; current manual documents the gap explicitly.

Encryption / TPM

Hardware manual recommends **not** re-enabling ZFS encryption until TPM + recovery strategy is documented. IndianaDell rebuild does not touch encryption.

ZFS boot import

The installed host **must** keep force import enabled:

```
# /etc/default/zfs
ZPOOL_IMPORT_OPTS="-f"
```

`bin/rebuild-machine` does **not** write this file. After reinstall or recovery, verify it yourself (Chapter 3 checklist). Without `-f`, boot can hang when pools need force-import (post-export hostid mismatch, unclean shutdown).

When something fails

1. Read `scripts/rebuild/last-run.log`
2. Run `bin/rebuild-machine --verify-only` for targeted MISS lines

3. Re-run individual phases (apt, HackRF build, Mayhem download) manually
4. Check chapter-specific verify sections

Reporting issues

Capture:

```
bin/dellmerge > debug.report
bin/rebuild-machine --verify-only 2>&1 | tee verify.log
uname -a && lsb_release -a
```

Chapter 15 — Ventoy Live Session & Persistence

This chapter covers **two** Ventoy volumes used on the lab:

Volume	Friendly name	Role	Typical size
Internal Seagate ST500DM002	Uncle Wiggly ☐☐	Full rabbit hole: many ISOs, 24 GB Ubuntu persis- tence, DOS- BOOT / Win- dows / ISO- STASH	~466 GB disk
PNY USB 2.0 (~30 GB)	Rebuild stick	Lean Ubuntu 26.04 live + 3 GB per- sistence for recovery / thumper rebuild	~30 GB USB

Both use Ventoy: drop ISOs onto the data partition; they appear in the boot menu. Writable **casper-rw** persistence keeps login state across live boots.

EFI note (Tower5810): Uncle Wiggly's SATA port may be **disabled in Setup** for fast POST (docs/fast-boot.md). If /dev/disk/by-label/Wiggly is missing but the Seagate is cabled, re-enable that SATA port in **F2 Setup**, then reboot.

Overlay layout (both volumes)

Ubuntu live + Ventoy persistence uses an ext4 file labeled **casper-rw**:

```
persistence/ubuntu-26.04.dat    # ext4, LABEL=casper-rw
  upper/                        # modern overlay upper (preferred)
    home/<liveuser>/
    etc/
    usr/local/sbin/
    ...
  work/                          # created by live system as needed
```

Older seeds may have used cow/upper/; current IndianaDell seeds use **upper/**. Seed scripts prefer upper/ and remove a stale cow/ tree when seeding.

Ventoy config (on the same data partition as the ISO):

```
{
  "persistence": [
    {
      "image": "/ubuntu-26.04-desktop-amd64.iso",
      "backend": "/persistence/ubuntu-26.04.dat",
      "autosel": 1
    }
  ]
}
```

Canonical repo copy for Uncle Wiggly: scripts/ventoy/ventoy.json. Paths are **absolute from the Ventoy data partition root**. Persistence backend and ISO must live on the **same** Ventoy data partition.

Subdirectories and non-ISO files

Ventoy **recursively scans** the data partition for bootable images (.iso, etc.). Subfolders are fine and already used on Wiggly:

Path example	Notes
/perc/*.iso	PERC FreeDOS/Linux kit; FreeDOS can use auto_memdisk in ventoy.json
/Windows NT ISOs/*.iso	Period OS media
/QubesOS/Qubes-....iso + .asc + signing key	Signatures sit next to the ISO; Ventoy ignores non-bootable files

Optional: empty .ventoyignore in a folder hides that tree from the boot menu.

Create a persistence image (example 3 GiB):

```
# From extracted Ventoy release, or scripts/ventoy/ if present
sudo ./CreatePersistentImg.sh -s 3072 -t ext4 -l casper-rw -o persistence/ubuntu-26.04.dat
# Extend later:
sudo scripts/ventoy/ExtendPersistentImg.sh /path/to/ubuntu-26.04.dat <extra-MB>
# then resize2fs on the loop device if needed
```

Uncle Wiggly ☐☐ (internal Ventoy)

Names: friendly **Uncle Wiggly**; partition label **Wiggly** (historically sdc1, mount /mnt/wiggly). Same Seagate also holds **DOSBOOT**, Windows, **ISO-STASH** — see hardware inventory.

What gets persisted (full seed)

Item	Live path	In casper image
User home	/home/ubuntu (Wiggly full seed convention)	upper/home/ubuntu/
Installed packages	dpkg overlay	upper/var/lib/dpkg/
GDM autologin	/etc/gdm3/custom.conf	upper/etc/gdm3/
Grok auth + sessions	~/.grok/	same (never in git)
GitHub CLI auth	~/.config/gh/	same
SSH keys	~/.ssh/	same

Item	Live path	In casper image
Chrome (tier C) Runtime secret source	~/ .config/google-chrome/ curated /home/user/ when ZFS rpool is available	bookmarks, prefs, logins, Web Data, Extensions — no caches pulled via resolve-secrets.sh
IndianaDell workspace PATH overrides	~/Documents/IndianaDell workspace ~/ .config/indianadell/path.sh	Full tree (git clone or rsync)

Persistence image: /persistence/ubuntu-26.04.dat (**24 GB** ext4, label casper - rw) on label **Wiggly**.

One-time setup (Tower5810)

```
sudo mount -o uid=$(id -u),gid=$(id -g) /dev/disk/by-label/Wiggly /mnt/wiggly
bin/setup-wiggly-ventoy # verify ISO, ventoy.json, .dat filesystem
```

Seed session state

```
~/bin/seed-ventoy-persistence.sh
# or:
PERSIST_MOUNT=/mnt/persist-check ~/bin/seed-ventoy-persistence.sh
SEED_CHROME=c ~/bin/seed-ventoy-persistence.sh # default chrome tier
SEED_CHROME=off ~/bin/seed-ventoy-persistence.sh
```

Mode	When	Network?
Live casper overlay	Already booted from Ventoy persistence	No — local rsync only
External .dat seed	Seeding from Tower5810 / mounted volume	Only if IndianaDell must be git cloned

Network check: waits up to SEED_NETWORK_WAIT_SECS (default **120s**). Skip: SEED_SKIP_NETWORK_CHECK=1.

Chrome profile seed (SEED_CHROME)

Default **c**. Prefer /home/user/.config/google-chrome when rpool home exists; never copies Cache / Code Cache / GPU* / Service Worker.

Tier	What is copied
off / 0	Nothing
a	Bookmarks + Preferences
b	a + Local State + Secure Preferences
c	b + Login Data + Web Data + Extensions + Local Extension Settings
d	Reserved (same as c for now)

Login experience (Wiggly full seed)

1. **GDM autologin** — live user **ubuntu** (Wiggly seed convention; see rebuild stick for **user**)
2. **PATH** — IndianaDell bin/ and scripts/ via ~/.config/indianadell/path.sh
3. **Grok autostart** — often disabled (X-GNOME-Autostart-enabled=false)
4. **Installer** — no autostart; Desktop Install icon when needed

resolve-secrets.sh materializes secrets from /home/user when rpool exists, else Ventoy \$HOME.

PNY rebuild stick (lean USB)

Device: ~30 GB PNY USB (model often USB 2.0 FD). **Not** production storage — install/recovery only.

Why repartition / reinstall Ventoy

The old layout was ~6 GB Ventoy + 32 MB VTOYEFI + ~24 GB DOSBOOT. Ubuntu 26.04 desktop ISO is ~**6.1 GB**, so the small Ventoy slice could not hold ISO + persistence. Rebuild (2026-07) used Ventoy **1.1.16** force-install so almost the whole stick is one data partition:

Partition	Size	Label	Role
...1	~29.9 G	Ventoy	ISO + persistence + ventoy.json
...2	32 M	VTOYEFI	Ventoy EFI

```
# SAFETY: confirm USB transport + ~30 GB before -I
lsblk -o NAME,SIZE,MODEL,TRAN,LABEL
# From extracted ventoy-*-linux:
sudo bash ./Ventoy2Disk.sh -I -L Ventoy /dev/sdX # double-confirm y/y
```

Hiren's / old Ubuntu 22 / DOSBOOT content on that stick was wiped by reinstall. BartPE and current media live on **Uncle Wiggly** (or ISO-STASH).

Layout on the stick

```
/ubuntu-26.04-desktop-amd64.iso      # ~6.1 G (copied from Wiggly)
/persistence/ubuntu-26.04.dat        # 3 G casper-rw
/ventoy/ventoy.json                  # persistence map, autosel=1
```

Capacity rule of thumb: 6.1 G ISO + 3 G .dat ≈ 9 G used; leave free space for future ISOs. A **24 GB** Wiggly-class overlay does **not** fit on this stick.

Lean seed policy (rebuild stick)

Include (as live user home — see identity below):

Item	Source (Tower5810)	Notes
SSH keys	/home/user/.ssh/	id_rsa, config, authorized_keys; mode 700/600
GitHub CLI	/home/user/.config/gh/	hosts.yml token
Grok	/home/user/.grok/	auth + sessions; prune downloads/ to save space optional
GnuPG	/home/user/.gnupg/	
gitconfig	/home/user/.gitconfig	
Cursor auth	~/.config/cursor/auth.json	if present
Shell dots	.bashrc, .profile, ...	
IndianaDell config	~/.config/indianadell, autostart, dconf	
~/bin	host ~/bin helpers	
Keyrings	~/.local/share/keyrings	
Project PDFs only	Documents/IndianaDell/B1GMB42-*.pdf	manuals / trifolds / ZFS recovery — not full Documents tree

Exclude: full Documents/, .cache, Chrome bulk, cargo/rustup/wine/googleearth, full .local (except keyrings).

Seed path in the image: upper/home/user/... (after username change below).

Mount / edit / unmount pattern:

```
DAT=/run/media/$USER/Ventoy/persistence/ubuntu-26.04.dat # or by-label Ventoy
MOUNT=/mnt/persist-usb
sudo mkdir -p "$MOUNT"
LOOP=$(sudo losetup --show -f "$DAT")
sudo mount "$LOOP" "$MOUNT"
# edit $MOUNT/upper/ ...
sync
sudo umount "$MOUNT"
sudo losetup -d "$LOOP"
```

Live user identity: user (not ubuntu)

Casper creates the live account from `/etc/casper.conf`. **FLAVOUR must be non-empty** or casper ignores USERNAME / HOST and substitutes the flavour string.

Seeded on the rebuild stick:

```
# /etc/casper.conf (in upper/)
export USERNAME="user"
export USERFULLNAME="User"
export HOST="thumper"
export BUILD_SYSTEM="Ubuntu"
export FLAVOUR="Ubuntu" # required for USERNAME/HOST to stick
```

Also:

File	Purpose
upper/etc/gdm3/custom.conf	AutomaticLoginEnable=true, AutomaticLogin=user
upper/etc/sudoers.d/casper	casper ALL=(ALL) NOPASSWD: ALL
upper/home/user/	Seeded secrets + lean home (uid/gid 1000)

Casper's `15autologin` / `25adduser` / `44pk_allow_ubuntu` all use `$USERNAME`, so PolicyKit and autologin follow `casper.conf` when `FLAVOUR` is set.

Boot unit: groups + rename safety net

Script: `upper/usr/local/sbin/indianadell-live-user.sh`

Unit: `indianadell-live-user.service` (WantedBy `multi-user.target` and `graphical.target`)

Log: `/var/log/indianadell-live-user.log`

Behavior:

1. If account **ubuntu** exists and **user** does not → `groupmod`/`usermod` rename + move home (old overlays / missed `FLAVOUR`).

2. Add **user** to every **special group that exists** on the live system (skip missing groups quietly).
3. Refresh sudoers + GDM autologin for user.
4. Set hostname **thumper** and `127.0.1.1 thumper.local thumper` in `/etc/hosts`.

Group list (lab / rebuild relevant — install packages later may create more groups; reboot re-runs attach):

Category	Groups
Admin / desktop	adm, cdrom, sudo, dip, plugdev, lpadmin, sambashare, users
Serial / radio	dialout, tty, uucp
AV / GPU / input	audio, video, render, input
Print / scan	lp, scanner
Network	netdev, bluetooth
Containers / VMs	docker, lxd, kvm, libvirt, libvirt-qemu
Other	disk, floppy, ssl-cert, wireshark, fuse

Matches the intent of Tower5810's host user groups (`id user`) plus hardware-facing groups we may install on live.

Fast-boot deferments (Tower5810 parity + live snap kill)

Same idea as `bin/apply-fast-boot/docs/fast-boot.md`, adapted for **casper overlay** (cannot run `systemctl disable` on the ISO lower layer — use **mask symlinks**).

Mechanism	Path in overlay	Role
Boot masks	<code>etc/systemd/system/<unit> → /dev/null</code>	Unit cannot start early (present as soon as root mounts)
Permanent mask list	<code>etc/indianadell-live-mask.list</code>	Never auto-start (snap seed, NM-wait-online, ...)

Mechanism	Path in overlay	Role
Deferred list	etc/indianadell-deferred.list	Unmasked + start --no-block after graphical.target
Socket-lazy list	etc/indianadell-socket-lazy.list	docker / cups / snapd / libvirt sockets only
Early assert	indianadell-live-fastboot.service □ usr/local/sbin/indianadell-live-fastboot.sh	Re-mask/stop thrashers; log /var/log/indianadell-live-fastboot.log
Post-desktop start	indianadell-deferred.service □ usr/local/sbin/indianadell-start-deferred	After GDM; does not block desktop

Permanent masks (rebuild stick — the usual multi-minute hang):

- snapd.seeded.service (main culprit on Ubuntu live)
- snapd.autoimport.service, snapd.core-fixup.service, snapd.recovery-chooser-trigger.service, ...
- NetworkManager-wait-online.service

Deferred (masked at boot, started later in background): snapd.service, cloud-init, cups, bluetooth, avahi, ModemManager, docker/libvirt stack, apport/whoopsie, unattended-upgrades, etc. (full list on the stick).

snapd.service itself is only started after the desktop is up (and only if you unmask/start it); **seeded** stays masked so it never blocks login.

Verify on live:

```
systemctl is-enabled snapd.seeded.service    # expect: masked
systemctl is-active snapd.seeded.service     # inactive
systemd-analyze blame | head -20
sudo tail /var/log/indianadell-live-fastboot.log
```

Plymouth theme (indianadell)

Tower's custom theme is seeded into the overlay (~20 MB):

Item	Overlay path
Theme tree	usr/share/plymouth/themes/indianadell/ (from host /usr/share/plymouth/themes/indianadell)
Default alternative	etc/alternatives/default.plymouth □ .../indianadell.plymouth
Default symlink	usr/share/plymouth/themes/default.plymouth □ alternatives
Daemon config	etc/plymouth/plymouthd.conf □ Theme=indianadell

indianadell-live-fastboot.sh re-asserts the default theme once rootfs is up.

Caveat: The **very first** splash frames still come from the **ISO initrd** (stock spinner/BGRT). After casper mounts the persistence overlay, Plymouth uses Theme=indianadell for late boot / shutdown / any restart of plymouthd. Fully replacing early ISO splash would require rebuilding casper/initrd on the ISO (not done on this stick).

Install source on Tower: `sudo bin/themes-install-boot` (see Ch. 5). Stick seed copies the **installed** theme tree, not a separate rebuild of animation frames.

Hostname thumper / thumper.local and OpenSSH on LAN

Hostname: casper.conf HOST=thumper plus /etc/hostname and /etc/hosts (thumper.local). Cloud-init can be steered with upper/etc/cloud/cloud.cfg.d/99-thumper-hostname.cfg if present.

SSH unit (optional seed): thumper-lan-ssh.service □ upper/usr/local/sbin/thumper-lan-ssh.sh

- Waits for network if needed; installs **openssh-server** (apt, or debs under /var/cache/thumper-debs/ if cached).
- Drop-in /etc/ssh/sshd_config.d/99-thumper-lan.conf: listen 0.0.0.0 / ::, pubkey + password auth, no root login, no empty passwords.
- Enables/starts ssh.service (disables socket-only activation if needed).
- Allows UFW OpenSSH if UFW is active; starts avahi-daemon when available for mDNS **thumper.local**.
- Log: /var/log/thumper-lan-ssh.log

Login:

```
ssh user@thumper.local  
# or LAN IP from the live session
```

Use the same SSH private key as Tower5810 (id_rsa); public key is in ~/.ssh/authorized_keys on the stick. Live password is typically blank at the console (casper); SSH should use **keys** (empty passwords disabled for SSH).

MOTD hint (if seeded): etc/update-motd.d/99-thumper-ssh.

Secrets policy (resolve-secrets.sh)

Canonical secret relative paths (never commit to git):

```
.ssh  
.grok  
.config/gh
```

When ZFS rpool /home/user is present, it is the live secret **source**; Ventoy persistence is the portable **store** under the live home. Chrome seed is separate (SEED_CHROME).

ZFS recovery (rpool + bpool)

Manual: docs/B1GMB42-zfs-recovery.md + B1GMB42-zfs-recovery.pdf (repo root and DOSBOOT recovery kit).

Boot Ventoy Ubuntu live — **do not** use a broken installed system as root.

```
sudo apt-get install -y zfsutils-linux  
cd ~/Documents/IndianaDell # or recovery kit path  
sudo ./mount-rpool-recovery.sh mount  
sudo ./scripts/recovery/mount-bpool-recovery.sh mount  
sudo ./mount-rpool-recovery.sh chroot  
# repair; then exit and umount both scripts
```

Before rebooting installed OS: /etc/default/zfs must set ZPOOL_IMPORT_OPTS="-f". Kernel one-shot: zfsforce=1.

Deploy kit: bin/deploy-dosboot-recovery (from Tower5810). PDF on the rebuild stick: ~/Documents/IndianaDell/B1GMB42-zfs-recovery.pdf.

GitHub repository

<https://github.com/webaugur/IndianaDell> (private)

```
bin/pull-repo --verify
bin/pull-repo --dragonsdr
bin/push-repo
```

HTTPS optional: `INDIANADELL_REMOTE=...` after `gh auth login`. Large FactoryDocs use **Git LFS**.

On the rebuild stick, after network is up: `git clone git@github.com:webaugur/IndianaDell.git` using seeded SSH keys.

How to verify

Uncle Wiggly (full)

```
findmnt / | grep -qE 'cow|overlay' && echo "persistence overlay active"
grep AutomaticLogin= /etc/gdm3/custom.conf
echo "$INDIANADELL_ROOT"
which dellmerge pull-repo push-repo grok
bin/pull-repo --verify
```

PNY rebuild stick

Identity

```
whoami # expect: user
id # expect sudo,adm,plugdev,dialout,... as available
hostname; hostname -f # thumper / thumper.local
grep AutomaticLogin=user /etc/gdm3/custom.conf
```

Persistence

```
findmnt / | grep -qE 'cow|overlay' && echo "overlay ok"
test -f ~/.ssh/id_rsa && test -f ~/.config/gh/hosts.yml && test -f ~/.grok/auth.j
ls ~/Documents/IndianaDell/*.pdf
```

SSH on LAN (after thumper-lan-ssh has run)

```
ss -tlnp | grep ':22'
systemctl is-active ssh
# from another host:
# ssh user@thumper.local
```

Fast-boot / snapd not thrashing

```
systemctl is-enabled snapd.seeded.service # masked
systemd-analyze blame | head -15
```

```
# Plymouth (rootfs theme)
grep Theme= /etc/plymouth/plymouthd.conf # indianadell
readlink -f /etc/alternatives/default.plymouth

# Logs
sudo tail -50 /var/log/indianadell-live-user.log
sudo tail -50 /var/log/thumper-lan-ssh.log
sudo tail -50 /var/log/indianadell-live-fastboot.log
```

How to customize

Goal	Action
Re-seed Wiggly full session	~/bin/seed-ventoy-persistence.sh
Change Grok session	Edit GROK_SESSION_ID in grok-indianadell-launch.sh
Enlarge persistence	ExtendPersistentImg.sh + resize2fs
Verify Wiggly layout	bin/setup-wiggly-ventoy
Edit rebuild stick overlay	loop-mount ubuntu-26.04.dat, edit upper/, unmount
Live user name	upper/etc/casper.conf (USERNAME + FLAVOUR) + GDM + home dir + indianadell-live-user.sh
Extra groups	Edit SPECIAL_GROUPS in indianadell-live-user.sh
SSH / hostname	thumper-lan-ssh.sh + sshd drop-in + hosts/hostname
Boot defer / mask lists	etc/indianadell-deferred.list, etc/indianadell-live-mask.list
Refresh Plymouth on stick	rsync host /usr/share/plymouth/themes/indianadell/ into overlay; keep plymouthd.conf
Hide ISO folder from menu	.ventoyignore in that folder
Qubes + signatures	e.g. Wiggly/QubesOS/*.iso + .asc + key (subdirs OK)

Related tools

Tool	Role
bin/setup-wiggly-ventoy	Verify Wiggly ISO + ventoy.json + .dat
bin/boot-uncle-wiggly-vm	QEMU live + persistence smoke test
bin/boxes-import-wiggly-isos	GNOME Boxes domain per ISO on Wiggly
bin/setup-perc-ventoy	PERC FreeDOS/IT kit on Wiggly (perc/)
~/bin/seed-ventoy-persistence.sh	Full seed into casper image
scripts/ventoy/resolve-secrets.sh	Secret path policy + Chrome tiers
scripts/ventoy/ventoy.json	Canonical Wiggly Ventoy plugin config
Ventoy release	Build empty casper-rw .dat
CreatePersistentImg.sh	

What rebuild does / does not do

Does	Does not
Install Chrome, gh, git-lfs when run on a full live session	Auto-configure Ventoy ventoy.json
Document seed / stick layout in this chapter	Auto-run seed every boot (except stick units: groups/SSH) Manage Seagate partition map (Wiggly/DOSBOOT/Windows) Fit a 24 GB Wiggly .dat on the 30 GB PNY stick

Chapter 16 — QEMU

QEMU is the core emulation and virtualization engine used throughout the IndianaDell / DragonSDR lab. On Tower5810 it provides the full range of system emulation targets, KVM-accelerated guests, and the shared-library builds required by Velxio / picsimlab for ESP32 development.

The complete upstream documentation lives in the QEMU source tree under docs/. This chapter gives a high-level map of that documentation so you can quickly locate the relevant sections.

Sub-chapters (QEMU docs tree)

About

High-level information about QEMU itself:

- Supported build platforms and minimum requirements
- Emulation capabilities and architecture coverage
- Deprecated and removed features
- License and contribution overview

Devel

Developer and internals documentation (most relevant when working on QEMU itself or debugging deep issues):

- Build system, Kconfig, and module architecture
- TCG (Tiny Code Generator) internals and plugins
- QAPI/QOM code generation, memory model, RCU, atomics
- Block layer, migration, multi-threaded TCG, iothreads
- Secure coding practices, style guide, patch submission process
- Tracing, replay, and record/replay infrastructure

Interop

Interoperability specifications and external interfaces:

- QEMU Machine Protocol (QMP)
- Guest agent protocol
- Block replication, COLO (COarse-grained LOck-stepping) fault tolerance
- NVDIMM, memory hotplug, PCI expander bridges, SR-IOV, etc.

Specs

Hardware and firmware specifications that QEMU emulates or interacts with:

- ACPI, SMBIOS, device tree fragments
- Virtio, vhost, and paravirtualized device specifications
- Firmware and boot interface details

Building QEMU on Tower5810

The DragonSDR build script `tools/emulators/qemu-lcgamboa/build-all.sh` produces both:

- The special Velxio/ESP32-compatible shared libraries (`libqemu-xtensa.so`, `libqemu-riscv32.so`)

- Full-featured shared libraries for all other architectures with PipeWire, KVM, virglrenderer, Spice, vhost, and modern storage/networking support enabled by default.

See the script and its pinned commit for the exact feature set.

Further reading

For the absolute latest and most detailed information, always consult the docs/ directory inside the QEMU source tree you are building. The structure described above is stable across recent QEMU releases.

Appendix A — bin/ Launchers

All launchers live in ~/Documents/IndianaDell/bin/. **PATH** is set automatically via ~/.config/indianadell/path.sh (IndianaDell tools override system binaries).

Launcher	Runs	Chapter
rebuild-machine	scripts/rebuild2rebuild-machine.sh	
build-software-manual	scripts/docs/build-software-manual.sh	
build-all-docs	scripts/docs/build-all-docs.sh	
pull-repo	scripts/github/pull-all.sh — IndianaDell + LFS	
install-dragonsdr	(--dragonsdr optional) ~/Documents/DragonSDR/bin/install-suite — full SDR suite	8
push-repo	bin/push-repo □ GitHub webaurgur/IndianaDell (SSH default)	15

Launcher	Runs	Chapter
setup-wiggly-ventoy	scripts/ventoy/setup-wiggly-ventoy.sh — Uncle Wiggly ISO + ventoy.json + .dat	15
setup-perc-ventoy	scripts/perc/setup-perc-ventoy.sh — H710 FreeDOS/IT kit on Uncle Wiggly	hardware / PERC doc
boot-uncle-wiggly-vm	scripts/ventoy/boot-uncle-wiggly-vm.sh — QEMU live+persistencetest	15
boxes-import-wiggly-isos	scripts/ventoy/boxes-import-wiggly-isos.sh — Boxes VM per ISO on Wiggly	15
themes-preview-boot	Themes/scripts/plymouth-preview.py — safe Plymouth window	15
apply-fast-login	scripts/gnome/apply-fast-login.sh — GRUB 0s + GDM autologin + face	15
apply-fast-boot	scripts/gnome/apply-fast-boot.sh — strip crashkernel, defer daemons	15
build-zfs-recovery-doc	scripts/docs/build-zfs-recovery-doc.sh	15
build-trifold-slick	docs/sales/B1GMB42-trifold.html □ sales PDFs	15

Launcher	Runs	Chapter
deploy-dosboot-recovery	scripts/recovery/05deploy-to-dosboot.sh	2/05
efi-timing-suite	scripts/efi/efi6, 12 timing-suite.sh	6, 12
dellmerge	scripts/dell/dell2merge.sh	12
gpu-stress	scripts/gpu/gpu6, 12 stress.sh	6, 12
iotest	scripts/storage/12iotest.sh	12
apply-amdgpu	etc/apply.sh	6
apply-sensor-watch	bin/apply-sensor-watch — install thermal/fan watchdog unit	6
indiana-sensor-watch	scripts/sensors/08indiana-sensor-watch.sh — --once / daemon	8
indiana-monitor-input	scripts/display/08indiana-monitor-input.sh — DDC/CI input select	8
indiana-ir-send	scripts/ir/indiana-ir-send.sh — USB serial to Uno IR blaster	6
indiana-ir-flash	scripts/ir/indiana-ir-flash.sh — arduino-cli compile/upload	6
tsci / tscircuit	tscircuit CLI under ~/.local — TypeScript □ KiCad export	EDA / Ch. 3
amd-install	amd-radeon/install-all.sh	6
amd-preflight	amd-radeon/00-preflight.sh	6

Launcher	Runs	Chapter
amd-verify	amd-radeon/04-verify.sh	6
amd-uninstall	amd-radeon/uninstall.sh	6
apply-dark-mode	scripts/gnome/apply-dark-mode.sh	5
apply-max-performance	scripts/gnome/apply-max-performance.sh	7
fix-nautilus-desktop-launch	scripts/gnome/fix-nautilus-desktop-launch.sh	7
sync-desktop-icons	scripts/gnome/sync-desktop-icons.sh	7
themes-extract	Themes/scripts/extract-all.sh	5
themes-install-boot	Themes/scripts/install-boot-theme.sh	5
themes-restore-boot	Themes/scripts/install-boot-theme.sh --restore-stock	5
hackrf-env	sources/DragonSDR/hackrf/scripts/env.sh	10
hackrf-setup-udev	DragonSDR/bin/hackrf-setup-udev	10
hackrf-download-mayhem	DragonSDR/bin/hackrf-download-mayhem	10
hackrf-prepare-sdcard	DragonSDR/bin/hackrf-prepare-sdcard	10
hackrf-flash-mayhem	DragonSDR/bin/hackrf-flash-mayhem	10
hackrf-build-mayhem	DragonSDR/bin/hackrf-build-mayhem	10

Ventoy session (scripts/ventoy/ □ ~/bin via install-ventoy-session.sh):

Script	Purpose
seed-ventoy-persistence.sh	Snapshot session into Ventoy casper image (Uncle Wiggly full seed)
seed-network-check.sh	Internet/DNS check before seed
grok-indianadell-launch.sh	Seed then Grok fullscreen autostart
install-ventoy-session.sh	Install helpers, autostart, PATH
mount-rpool-recovery.sh	ZFS rpool chroot recovery (workspace root)
mount-bpool-recovery.sh	ZFS bpool mount at /recovery/boot (scripts/recovery/)

PNY rebuild stick (lean 3GB persistence, live user user, groups, thumper/SSH): no dedicated bin/ launcher yet — procedure and overlay paths are documented in **Chapter 15** (upper/usr/local/sbin/indianadell-live-user.sh, thumper-lan-ssh.sh).

Note: hackrf-env must be **sourced**, not executed: source bin/hackrf-env

Sudo required: apply-amdgpu, themes-install-boot, themes-restore-boot, iotest, hackrf-setup-udev (udev install), amd-install, efi-timing-suite, mount-rpool-recovery.sh, mount-bpool-recovery.sh

Appendix B — Apt Packages by Chapter

Workstation packages: scripts/rebuild/package-lists.sh (APT_CORE + APT_KICAD).

SDR / ham / HackRF packages: ~/Documents/DragonSDR/tools/package-lists.sh (APT_SDR, APT_HAM, APT_SDR_BUILD).

Install SDR suite: bin/install-dragonsdr DragonSDR bin/install-suite.

Full system snapshot: apt-full-manifest.txt (after rebuild).

SDR/ham filter snapshot: apt-hamradio-dev-manifest.txt.

KiCad opt-out: SKIP_KICAD=1 (skips PPA + APT_KICAD + tscircuit npm on rebuild and fix-indianadell).

Chapter 4 — Development (IndianaDell APT_CORE)

build-essential, cmake, pkg-config, git, curl, wget, unzip, python3-pip, python3-venv, python3-dev, python3-numpy, python3-scipy, python3-matplotlib, python3-yaml, python3-requests, python3-pyqt5, python3-psutil, libssl-dev, clang, llvm-dev, libclang-dev, libusb-1.0-0-dev, libfftw3-dev, libvolk-dev,

portaudio19-dev, libsndfile1-dev, libboost-dev, libboost-program-options-dev, pandoc, texlive-latex-recommended, texlive-fonts-recommended, texlive-xetex, gh

Chapter 6 — GPU and Display

vulkan-tools, mesa-utils, mesa-utils-bin, clinfo, x11-apps, smartmontools, ddcutil, arduino-cli

Chapter 8 — GNU Radio and SDR (DragonSDR APT_SDR + build libs)

gnuradio, gnuradio-dev, gnuradio-doc, gr-osmosdr, gr-limesdr, gr-fosphor, gr-air-modes, gr-hpsdr, gr-dab, gr-satellites, libsoapysdr-dev, python3-soapysdr, soapysdr-module-osmosdr, soapysdr-module-mirisdr, uhd-soapysdr, rtl-sdr, librtlsdr-dev, airspy, libairspy-dev, bladerf, libbladerf-dev, limesuite, limesuite-udev, uhd-host, libuhd-dev, gqrx-sdr, quisk, inspectrum, hacktv

Chapter 9 — Ham Radio (DragonSDR APT_HAM)

libhamlib-dev, libhamlib-utils, python3-hamlib, fldigi, wsjtx, wsjtx-data, chirp, direwolf, gpredict, grig, xastir, xastir-data

Chapter 10 — HackRF and Mayhem (DragonSDR APT_SDR)

hackrf, hackrf-firmware, libhackrf-dev, hackrf-doc, dfu-util, openocd, gcc-arm-none-eabi, binutils-arm-none-eabi, libnewlib-arm-none-eabi, ccache, lz4, bzip2

EDA — KiCad 10 (APT_KICAD)

Default-on for workstation rebuild. Requires ppa:kicad/kicad-10.0-releases (scripts/rebuild/ensure-kicad-ppa.sh) before install so apt does not pull universe 9.x.

kicad, kicad-libraries, kicad-symbols, kicad-footprints, kicad-packages3d, kicad-templates, kicad-demos, kicad-dbg, kicad-doc-en, kicad-doc-de, kicad-doc-fr, kicad-doc-es, kicad-doc-it, kicad-doc-ja, kicad-doc-pl, kicad-doc-ru, kicad-doc-zh, kicad-doc-ca, kicad-gruvbox-theme, ngspice, gerbv, nodejs, npm

tscircuit (not apt): scripts/rebuild/install-tscircuit.sh runs npm install -g --prefix ~/.local tscircuit @tscircuit/capacity-autorouter typescript. CLI is tsci / tscircuit. Export a circuit

to KiCad with `tsci export circuit.tsx -f kicad_sch` (also `kicad_pcb`, `kicad_zip`). Autorouter npm name is `@tscircuit/capacity-autorouter` (`tscircuit-autorouter`).

`kicad-doc-id` is **not** in the list (universe still 9.0.8 while the PPA is 10.x). Skip the whole set (including `tscircuit`) with `SKIP_KICAD=1`.

Chapter 11 — Flatpak

flatpak (`application.org.telegram.desktop` installed via flatpak, not apt)

Chapters 5, 7, 12, 13 — No dedicated apt arrays

Themes, GNOME prefs, machine utilities, and FactoryDocs use workspace scripts or Ubuntu desktop packages already on the base install.